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UNIT 3 PROJECT

Sleep, Screen Time and Stress Analysis

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GitHub repository: Github link

Dashboard: Streamlit app link

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1. Introduction

1.1 Dataset

Table 1. *Dataset information*

Item	Details
Title	Sleep, Screen Time and Stress Analysis
Source	Kaggle (https://www.kaggle.com/datasets/jayjoshi37/sleep-screen-time-and-stress-analysis/)
Format	CSV file
Size	15,000 rows and 13 columns (14,415 rows after cleaning)

Description. This dataset explores the relationship between smartphone usage, sleep patterns and stress levels. Each row is one person, described by age, gender, occupation, daily screen time, phone use before sleep, sleep duration and quality, caffeine intake, physical activity, notifications per day, a stress level and a mental fatigue score. The people are adults aged 18 to 59 across eight occupations.

Why I selected it. Smartphones are part of almost everyone's daily life, and I wanted to explore whether heavy screen time is linked to poorer sleep and higher stress.

1.2 Problem definition

What is the dataset about? It describes how people use their phones and how they sleep, together with how stressed and mentally tired they are. The goal of this project is to find out which habits are most closely linked to stress.

Main questions.

- Is higher daily screen time associated with higher stress levels?
- Do people who sleep fewer hours tend to experience higher stress?
- Is using a smartphone before sleeping associated with shorter sleep duration or poorer sleep quality?
- How do caffeine consumption and physical activity relate to sleep and stress?
- Do stress levels vary across occupations, and do high-pressure jobs such as doctors have higher stress?
- Which variables have the strongest relationships with stress levels?
- Do certain occupations, such as doctors, or other high-pressure professions, tend to have higher stress levels than other occupations?

Expected insights. I expected a strong positive relationship between screen time and stress, some effect of caffeine and physical activity on sleep and stress, and higher stress in high-pressure occupations such as doctors and software engineers.

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1.3 Tools

The analysis was done in Python with pandas and NumPy for data handling, Matplotlib and Seaborn for the exploratory charts in a Jupyter notebook, and Plotly with Streamlit for the interactive dashboard.

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2. Data Exploration

The data was loaded with pandas (`df = pd.read_csv(...)`) and inspected with `head()`, `shape`, `info()`, `describe()` and `isnull().sum()`.

2.1 First rows and shape

The dataset has **15,000 rows and 13 columns** (`df.shape` returns (15000, 13)). Table 2 shows the first five rows; the columns are listed down the side to fit the page.

Table 2. *First five rows of the dataset (df.head())*

Column	Row 0	Row 1	Row 2	Row 3	Row 4
user_id	1	2	3	4	5
age	56	46	32	25	38
gender	Female	Female	Female	Male	Female
occupation	Designer	Teacher	Designer	Software Engineer	Teacher
daily_screen_time_hours	3.26	1.85	3.04	9	3.52
phone_usage_before_sleep_minutes	86	32	107	36	56
sleep_duration_hours	5.31	7.36	4.50	6.68	7.57
sleep_quality_score	7.72	9.70	6.38	5.53	6.69
stress_level	3.49	3.01	5.03	10	6.71
caffeine_intake_cups	0	0	0	0	4
physical_activity_minutes	35	16	17	3	92
notifications_received_per_day	119	299	21	220	167
mental_fatigue_score	3.57	1.91	6.05	9.92	5.99

2.2 Data types and missing values

Table 3. *Columns, data types and missing values (df.info() and df.isnull().sum())*

Column	Meaning	Data type	Missing
user_id	Unique number for each person	int64	0
age	Age in years	int64	0
gender	Female, Male or Other	object (text)	0
occupation	One of eight occupations	object (text)	0
daily_screen_time_hours	Daily screen time in hours	float64	0
phone_usage_before_sleep_minutes	Minutes of phone use before sleeping	int64	0
sleep_duration_hours	Hours slept per night	float64	0
sleep_quality_score	Sleep quality score (1 to 10)	float64	0
stress_level	Stress level (1 to 10)	float64	0
caffeine_intake_cups	Cups of caffeine per day	int64	0
physical_activity_minutes	Minutes of physical activity per day	int64	0
notifications_received_per_day	Phone notifications per day	int64	0
mental_fatigue_score	Mental fatigue score (1 to 10)	float64	0

All 13 columns have 15,000 non-null values, so there are **no missing values** (0 in total) and **no duplicate rows** (0). The numerical columns are stored as numbers, and gender and occupation are text categories.

2.3 Summary statistics

Table 4. *Summary statistics of the numerical columns (df.describe())*

Column	count	mean	std	min	25%	50%	75%	max
user_id	15,000	7500.50	4330.27	1.00	3750.75	7500.50	11250.25	15000.00
age	15,000	38.49	12.01	18.00	28.00	38.00	49.00	59.00
daily_screen_time_hours	15,000	5.50	2.60	1.00	3.26	5.49	7.76	10.00
phone_usage_before_sleep_minutes	15,000	59.71	34.64	0.00	29.00	60.00	90.00	119.00
sleep_duration_hours	15,000	6.51	1.45	4.00	5.26	6.49	7.79	9.00
sleep_quality_score	15,000	6.25	1.71	1.00	5.00	6.25	7.50	10.00
stress_level	15,000	6.98	2.75	1.00	4.75	7.38	10.00	10.00
caffeine_intake_cups	15,000	2.00	1.41	0.00	1.00	2.00	3.00	4.00
physical_activity_minutes	15,000	59.16	34.53	0.00	29.00	59.00	89.00	119.00
notifications_received_per_day	15,000	160.89	80.86	20.00	92.00	162.00	231.00	299.00
mental_fatigue_score	15,000	6.87	2.73	1.00	4.70	7.38	9.45	10.00

Key points from the statistics:

- Ages range from 18 to 59 with a mean of about 38.5 years.
- Average daily screen time is about 5.5 hours (range 1 to 10) and average sleep is about 6.5 hours (range 4 to 9).
- Stress level, sleep quality and mental fatigue are scores from 1 to 10. The average stress level is about 7, and the 75th percentile is already at the maximum of 10.
- Caffeine ranges from 0 to 4 cups a day, and phone use before sleep and physical activity both range from 0 to about 2 hours (119 minutes).

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3. Data Cleaning

The cleaning steps were chosen based on what the exploration showed. Table 5 summarises each step and its result.

Table 5. *Data cleaning steps*

Step	What was found	Action	Result
Missing values	No missing values in any column	None needed	No imputation required
Duplicates	No duplicated rows	None needed	No rows dropped
Invalid categories	Gender contains Female, Male and Other. Occupation has 8 valid values. 585 rows have gender "Other"	Removed rows with gender "Other", because there is nothing called gender 'Other'	15,000 rows reduced to 14,415
Data types	Numerical columns are already numeric; gender and occupation are text categories	No conversion required	Types kept as they are
Rename columns	notifications_received_per_day is long	Renamed to notifications_per_day	Shorter, easier to use
Outliers	Boxplots show only 9 outliers, all in sleep_quality_score	Checked with the IQR rule and kept (see below)	No rows dropped
Save	-	Saved the cleaned data to CSV	sleep_mobile_stress_dataset_15000_cleaned.csv

3.1 Outlier check

The boxplots in Section 4 show that sleep quality is the only variable with points outside the bounds. Using the IQR rule, Q1 = 5.00, Q3 = 7.49 and IQR = 2.49, so the lower bound is 1.26 and the upper bound is 11.23. 9 people fall below the lower bound, with sleep quality scores of about 1. These are valid values on a 1 to 10 scale, and these people also have the maximum stress (average 10) and very high mental fatigue (average 9.64). They are consistent with the rest of the data, so I kept them.

After cleaning, the dataset has **14,415 rows and 13 columns** and was saved as a CSV file, which is also the file used by the Streamlit dashboard.

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4. Exploratory Data Analysis (EDA)

The EDA starts with single variables (univariate analysis) and then looks at relationships between variables (bivariate and multivariate analysis). All charts use the cleaned data.

4.1 Univariate analysis

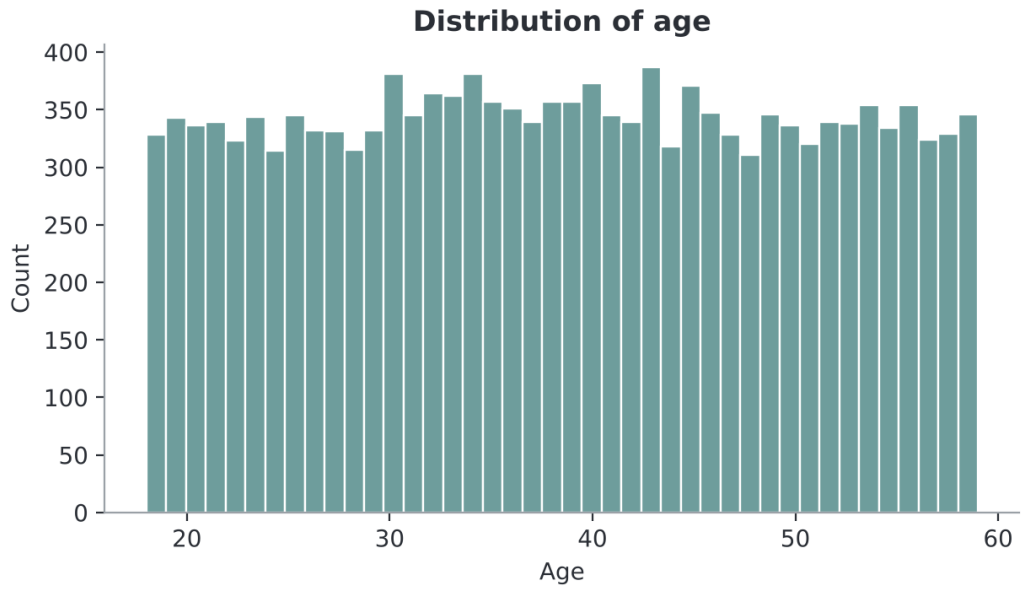


Figure 1. *Distribution of age with one bar per year. The counts are similar for every age from 18 to 59, so ages are roughly uniformly distributed.*

Figure 1 shows that there are about 310 to 390 people of every age. The distribution is flat, with no age group that dominates the dataset.

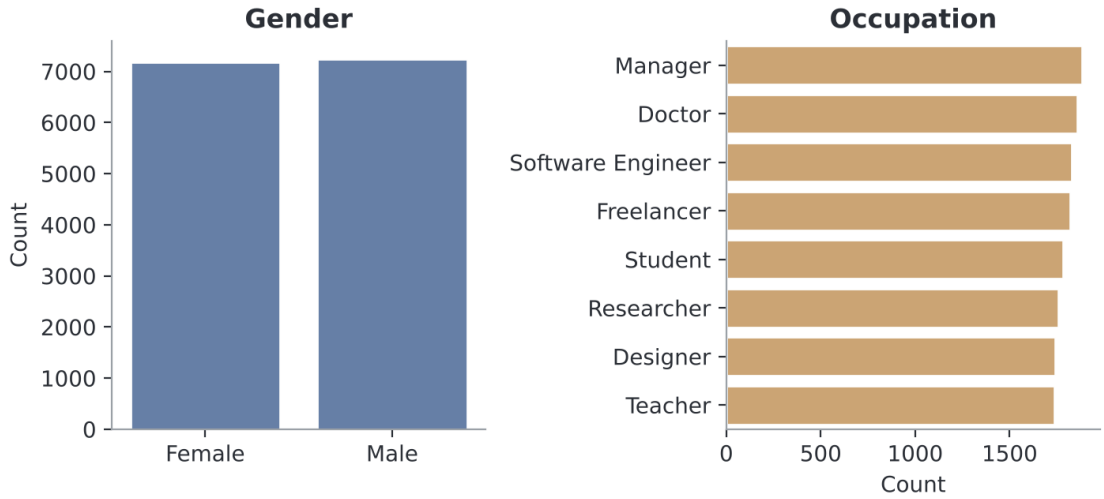


Figure 2. *Number of people by gender (left) and by occupation (right). Both variables are almost perfectly balanced.*

Gender is balanced (7,234 males and 7,181 females), and each of the eight occupations has between 1,738 and 1,885 people. Because the groups are balanced, comparisons between them are fair.

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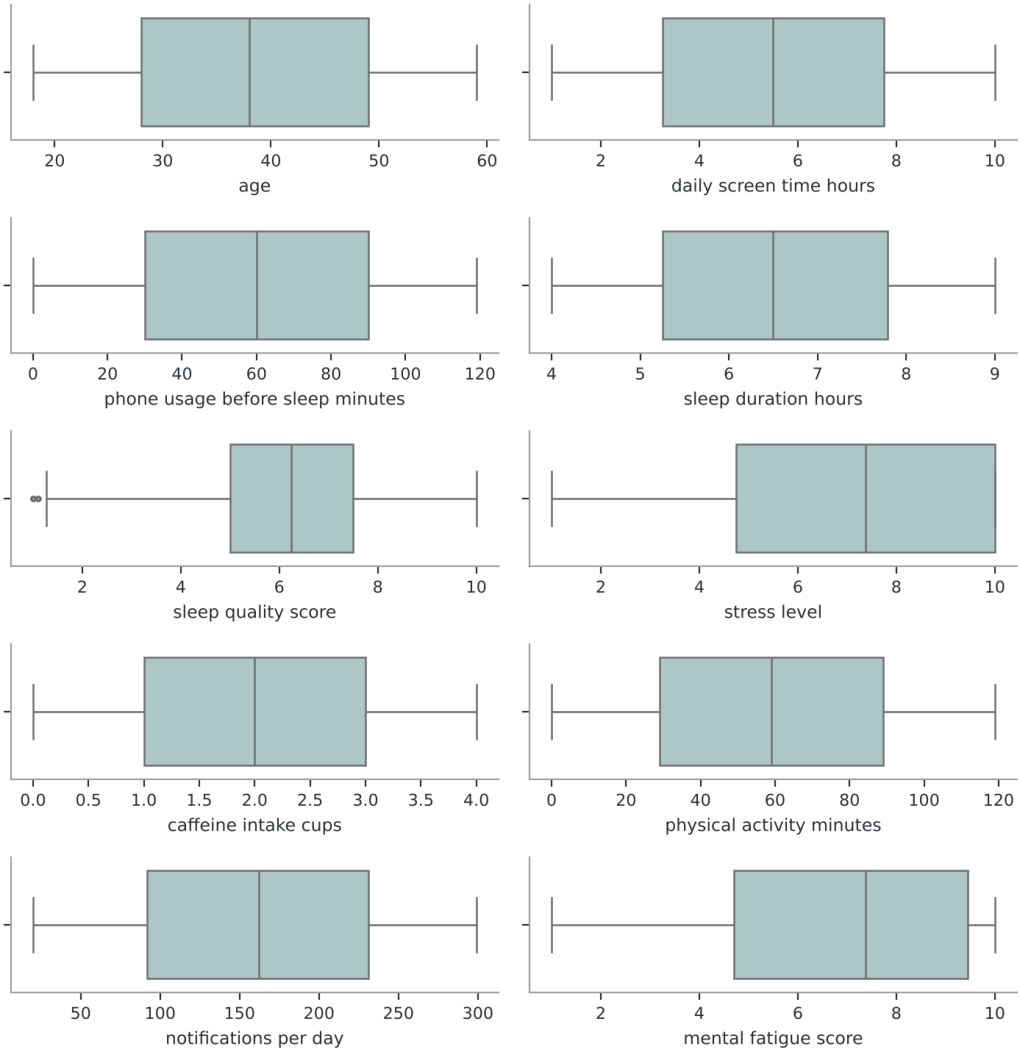


Figure 3. *Boxplots of all numerical variables. Sleep quality is the only variable with outliers (small points on the left).*

Most variables spread across their whole range. Stress level and mental fatigue are bunched toward the high end, with the median stress at about 7.4 and the upper quartile at the maximum score of 10.

4.2 Bivariate analysis

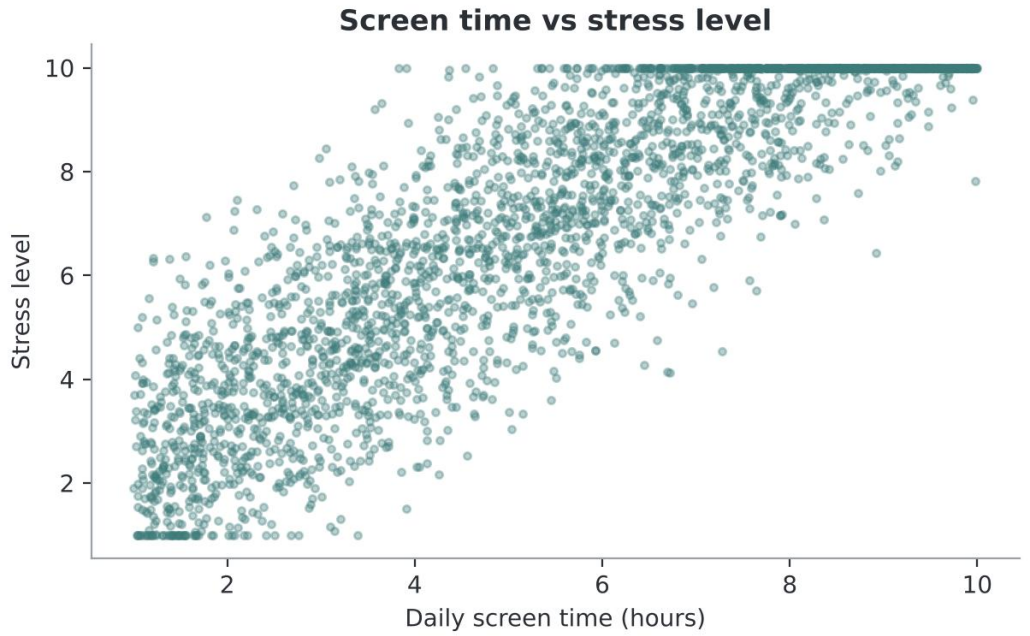


Figure 4. *Scatter plot of daily screen time against stress level (random sample of 3,000 people). Stress rises steadily with screen time.*

Figure 4 shows a clear upward trend: people with more screen time report higher stress (correlation 0.88). The row of points at the top shows that stress is capped at 10.

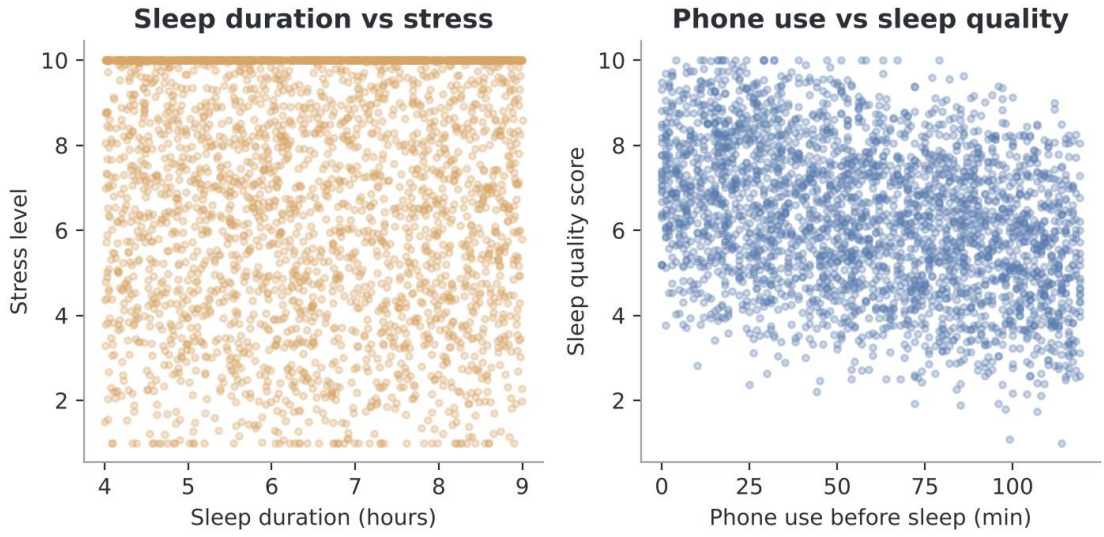


Figure 5. *Left: sleep duration against stress level, which shows no pattern. Right: phone use before sleep against sleep quality, which shows a downward trend.*

Sleeping more hours does not reduce stress (correlation 0.00), so the number of hours slept is not the key factor here. Phone use before sleep does not change sleep duration either, but it is linked to lower sleep quality (correlation -0.33): people who use their phone more before bed tend to sleep less well.

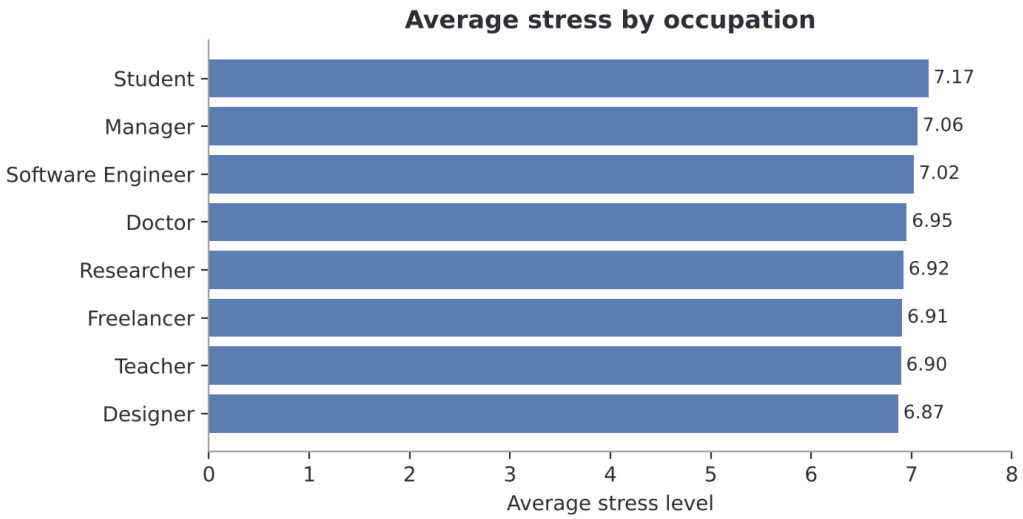


Figure 6. *Average stress level for each occupation. The differences are small, about 0.3 points between the highest and the lowest.*

Students have the highest average stress (7.17) and designers the lowest (6.87), but the gap is only 0.30 points. High-pressure jobs such as doctors (6.95) are not noticeably more stressed than other occupations.

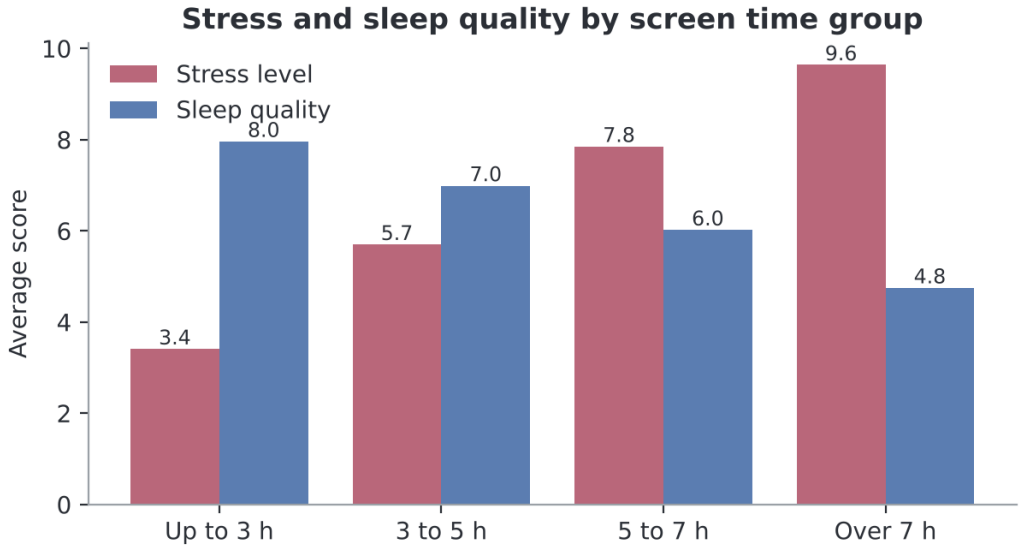


Figure 7. *Average stress level and sleep quality for four groups of daily screen time. More screen time goes with higher stress and lower sleep quality.*

When people are grouped by screen time, average stress rises from 3.41 (up to 3 hours) to 9.64 (over 7 hours), while sleep quality falls from 7.96 to 4.75.

4.3 Multivariate analysis

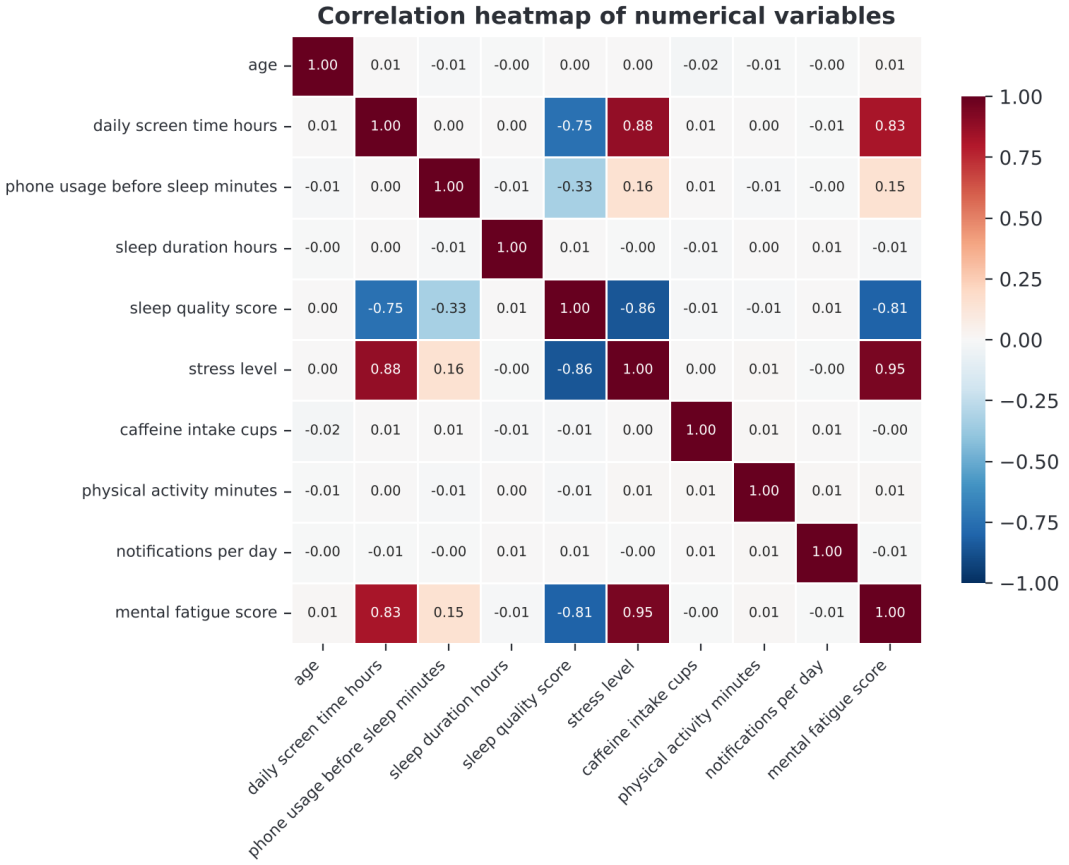


Figure 8. *Correlation heatmap of all numerical variables (user_id excluded). Red means a positive correlation and blue a negative one.*

The heatmap shows three groups of variables. Stress, screen time and mental fatigue are strongly positively correlated with each other, and sleep quality is strongly negatively correlated with all three. The remaining variables (age, sleep duration, caffeine, physical activity and notifications) have correlations close to zero with everything. Table 6 lists the correlation of each variable with stress.

Table 6. *Correlation of each variable with stress level*

Variable	Correlation with stress
mental_fatigue_score	0.95
daily_screen_time_hours	0.88
sleep_quality_score	-0.86
phone_usage_before_sleep_minutes	0.16
physical_activity_minutes	0.01
notifications_per_day	-0.01
caffeine_intake_cups	0.00
sleep_duration_hours	0.00
age	0.00

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5. Dashboard Overview

The interactive dashboard was built with Streamlit and Plotly (`app.py`) and reads the cleaned CSV file. It can be started with `streamlit run app.py`. A calm colour theme is set in `.streamlit/config.toml`.

5.1 Sidebar

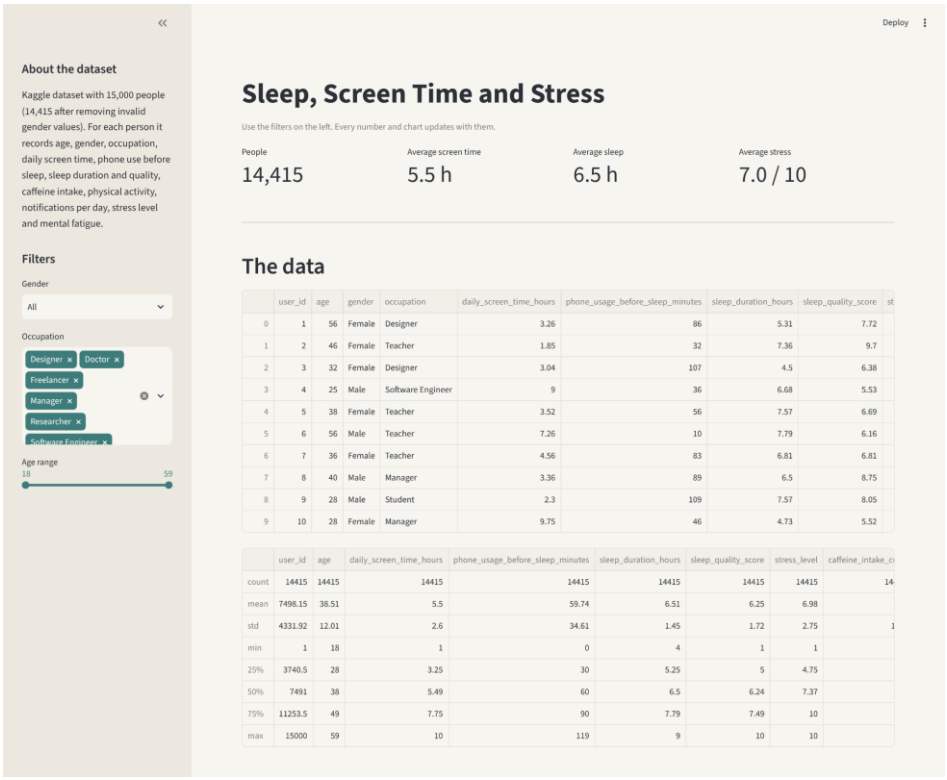
- **About the dataset:** a short description of the data.
- **Gender filter (dropdown):** All, Female or Male.
- **Occupation filter (multiselect):** choose any combination of the eight occupations.
- **Age filter (slider):** choose an age range between 18 and 59.

5.2 Main page

- **Summary numbers:** number of people, average screen time, average sleep and average stress.
- **Data preview and summary statistics:** the first rows of the filtered data and its describe() table.
- **Eight interactive charts:** hovering shows exact values. Every caption under a chart is calculated from the filtered data.
- **Key insights:** correlations that update when the filters change.

The filters apply to every number and chart at the same time. For example, selecting a single occupation or age range immediately recalculates all charts and insights. Figures 9 to 11 show the dashboard with all filters at their default values. An optional predictive model was not added.

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	user_id	age	gender	occupation	daily_screen_time_hours	phone_usage_before_sleep_minutes	sleep_duration_hours	sleep_quality_score	st
0	1	56	Female	Designer	3.26	86	5.31	7.72	
1	2	46	Female	Teacher	1.85	32	7.36	9.7	
2	3	32	Female	Designer	3.04	107	4.5	6.38	
3	4	25	Male	Software Engineer	9	36	6.68	5.53	
4	5	38	Female	Teacher	3.52	56	7.57	6.69	
5	6	56	Male	Teacher	7.26	10	7.79	6.16	
6	7	36	Female	Teacher	4.56	83	6.81	6.81	
7	8	40	Male	Manager	3.36	89	6.5	8.75	
8	9	28	Male	Student	2.3	109	7.57	8.05	
9	10	28	Female	Manager	9.75	46	4.73	5.52	

	user_id	age	daily_screen_time_hours	phone_usage_before_sleep_minutes	sleep_duration_hours	sleep_quality_score	stress_level	caffeine_intake_c
count	14415	14415	14415	14415	14415	14415	14415	14
mean	7498.15	38.51	5.5	59.74	6.51	6.25	6.98	
std	4331.92	12.01	2.6	34.61	1.45	1.72	2.75	1
min	1	18	1	0	4	1	1	
25%	3740.5	28	3.25	30	5.25	5	4.75	
50%	7491	38	5.49	60	6.5	6.24	7.37	
75%	11253.5	49	7.75	90	7.79	7.49	10	
max	15000	59	10	119	9	10	10	

Figure 9. *Dashboard: sidebar with the dataset description and filters, summary numbers, data preview and summary statistics.*



Figure 10. *Dashboard: average stress by screen time, correlation with stress, mental fatigue and sleep quality by screen time, and stress by occupation.*

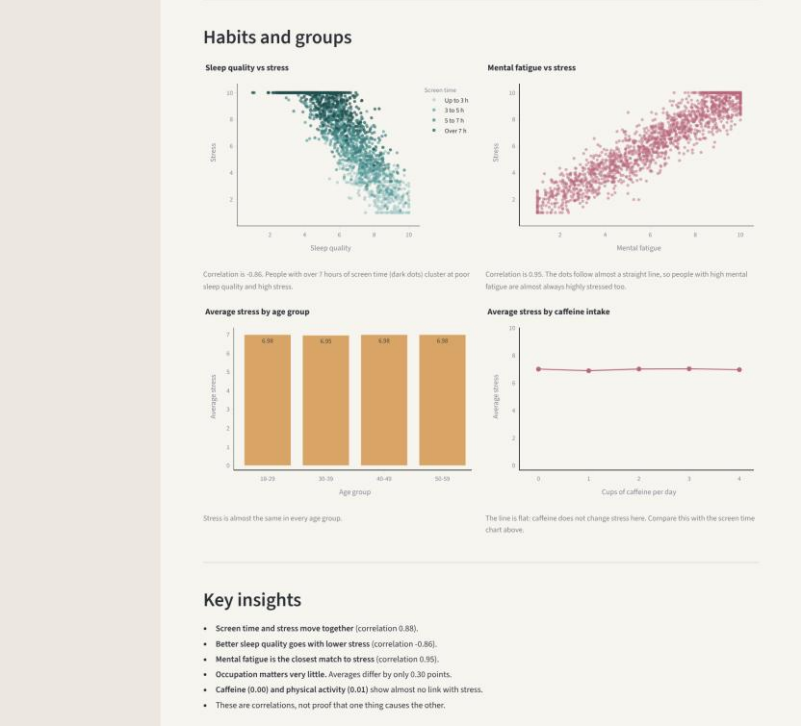


Figure 11. *Dashboard: sleep quality vs stress, mental fatigue vs stress, stress by age group, stress by caffeine intake, and the key insights section.*

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6. Insights and Conclusion

6.1 Insights summary

Patterns discovered

- Age is evenly spread from 18 to 59, and gender and occupation are almost perfectly balanced.
- Stress rises steadily with screen time: from 3.41 on average for people with up to 3 hours to 9.64 for people with over 7 hours.
- Stress levels are similar across occupations, ages and genders.

Interesting relationships

- Mental fatigue has the strongest relationship with stress (0.95), followed by screen time (0.88) and sleep quality (-0.86).
- Screen time is strongly linked to poorer sleep quality (-0.75), and phone use before sleep is moderately linked to it (-0.33).
- Sleep duration, caffeine intake and physical activity show almost no relationship with stress. What matters is how well people sleep, not how long.

Outliers

- Only 9 outliers were found, all in sleep quality (scores near 1). They belong to people with maximum stress and very high mental fatigue, so they were kept.

Key takeaways

- Reducing screen time and improving sleep quality appear to be far more important for stress than adjusting caffeine or physical activity in this dataset.
- Occupation does not explain stress in this data; habits do.

6.2 Answers to the main questions

Table 7. *Answers to the questions from Section 1.2*

Question	Answer
Screen time and stress	Yes. Strong positive relationship (0.88).
Sleep duration and stress	No. There is no relationship (0.00). Sleep quality matters instead (-0.86).
Phone use before sleep	Not linked to sleep duration, but linked to lower sleep quality (-0.33) and slightly higher stress (0.16).
Caffeine and physical activity	No relationship with sleep or stress (correlations of 0.01 or less).
Stress across occupations	Very little difference (6.87 to 7.17). Students are highest.
Strongest relationships	Mental fatigue (0.95), screen time (0.88) and sleep quality (-0.86).
High-pressure jobs	No. Doctors (6.95) and software engineers (7.02) are close to everyone else. There are no nurses in the dataset.

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6.3 What I learned

- How to follow a complete data analysis workflow: explore, clean, visualize, explain and present.
- That checking the data before cleaning matters, for example reading the same chart in more than one way (the first age histogram looked uneven only because of how the bins were drawn).
- That a strong correlation shows a link but does not prove a cause.
- How to turn an analysis into an interactive Streamlit dashboard with filters and charts that update together.

6.4 Limitations of the dataset

- **Source and realism.** The Kaggle page does not say how the data was collected. The flat age distribution, the almost perfect balance between groups and many correlations of exactly zero suggest that the data may be simulated, so the results may not describe real people.
- **Ceiling effect.** Scores stop at 10, and 25.8% of people have a stress level of 10. Real differences at the top are hidden, and correlations may be underestimated.
- **Correlation, not causation.** The data is a single snapshot, so it cannot show that screen time causes stress. Stress could also lead to more phone use.
- **Missing information.** There is no information about how scores were measured, whether they are self-reported, the type of screen use (work or leisure), health conditions or time of year.
- **Nurses.** The data has no nurses, so that question could only be answered for the occupations that exist.

6.5 Conclusion

In this dataset, stress is closely linked to screen time, poor sleep quality and mental fatigue, while age, gender, occupation, caffeine, physical activity and sleep duration have little or no relationship with it. The dashboard lets anyone repeat these checks for any age range, gender or occupation group. Because the data is correlational and may be simulated.

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7. References

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